THE LOCALISATION SEQUENCE OF CHOW GROUPS FOR AN OPEN SUBSPACE IS RIGHT EXACT

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a lifting of a rational equivalence from the open to the ambient space through generic points, a local finiteness check against quasi-compact opens, and an application of the exactness statement for the complement

Editorial note. The label lemma-restrict-to-open names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Rational equivalence by locally finite sums of divisors of rational functions on integral closed subspaces of one dimension higher, and restriction of cycles to an open subspace
- The reduced closed subspace structure on the complement of an open subspace and the closed immersion carrying it into the ambient space
- The exactness statement identifying cycles on the ambient space whose restriction to the open vanishes with the pushforwards of cycles on the closed complement, and local finiteness of a family of irreducible components in a Noetherian space

Source reference. Original reference 02RC: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 1306–1306. Prerequisite pointer invoked at proof line(s) 1987; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02QL: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 264–264. Prerequisite pointer invoked at proof line(s) 1952; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 02QL let X/B be good. Let $U \subset X$ be an open subspace. Let Y be the reduced closed subspace of X with $|Y| = |X| \setminus |U|$ and denote $i : Y \to X$ the inclusion morphism. Let $k \in \mathbf{Z}$. Suppose $\alpha, \beta \in Z_k(X)$. If $\alpha|_U \sim_{rat} \beta|_U$ then there exist a cycle $\gamma \in Z_k(Y)$ such that*

$$\alpha \sim_{rat} \beta + i_*\gamma.$$

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In other words, the sequence

$$\mathrm{CH}_k(Y) \xrightarrow{i_*} \mathrm{CH}_k(X) \xrightarrow{j^*} \mathrm{CH}_k(U) \longrightarrow 0$$

is an exact complex of abelian groups.

Proof. Let $\{W_j\}_{j\in J}$ be a locally finite collection of integral closed subspaces of U of δ-dimension $k+1$, and let $f_j \in R(W_j)^*$ be elements such that $(\alpha - \beta)|_U = \sum (i_j)_* \mathrm{div}(f_j)$ as in the definition. Let $W'_j \subset X$ be the corresponding integral closed subspace of X, i.e., having the same generic point as W_j. Suppose that $V \subset X$ is a quasi-compact open. Then also $V \cap U$ is quasi-compact open in U as V is Noetherian. Hence the set $\{j \in J \mid W_j \cap V \neq \emptyset\} = \{j \in J \mid W'_j \cap V \neq \emptyset\}$ is finite since $\{W_j\}$ is locally finite. In other words we see that $\{W'_j\}$ is also locally finite. Since $R(W_j) = R(W'_j)$ we see that

$$\alpha - \beta - \sum (i'_j)_* \mathrm{div}(f_j)$$

is a cycle on X whose restriction to U is zero. The lemma follows by applying Lemma 02RC. □